



Pneumonia Detection from Chest X-Rays

Classical ML vs Deep Learning — head-to-head with comparison tests

Module: Machine Learning · Track A: Computer Vision · Unit 11

[Student Name] · University of Essex Online · 2026

What I will cover

- 01 Design strategy — three-tier decision-provenance framework
- 02 Clinical motivation, dataset, EDA
- 03 Two models — HOG+SVM and ResNet50 transfer learning
- 04 Validation strategy and head-to-head evaluation
- 05 Explainability — HOG-viz + occlusion sensitivity (unified XAI)
- 06 Bias and subgroup diagnostics
- 07 Comparison tests — what worked best
- 08 Ethics, deployment, recommendation, reflections

Three-tier decision-provenance framework

Purpose: manage ML design uncertainty and make each key choice auditable.

TIER A — Direct research evidence

Peer-reviewed source directly supports the method or parameter.

Used when literature is specific enough.

E.g. HOG params (Dalal & Triggs 2005); Adam (Kingma & Ba 2014); early stopping (Prechelt 1998).

TIER B — Established practice / domain precedent

Official documentation, framework defaults, or strong prior work in a close domain.

Used when direct parameter-level evidence is absent.

E.g. ImageNet transfer (Yosinski 2014); CheXNet / Grad-CAM (Rajpurkar 2017); sklearn / TensorFlow defaults.

TIER C — Empirical project evidence

No reliable external prescription.

Run controlled comparison on this dataset.

E.g. HOG image size; CNN augmentation on / off.

Original project-level framework, informed by evidence-based-practice logic (Sackett et al., 1996) and design-science evaluation (Hevner et al., 2004). Critical caveats stated when evidence is indirect.

Pneumonia: leading cause of childhood mortality

740,000+

deaths under age 5 each year, worldwide (UNICEF)

- ML can triage where radiologists are scarce
- Flag urgent cases, not replace clinicians
- False negatives carry the dominant clinical cost
- Shortcut learning is a documented concern (Zech, 2018)

Chest X-Ray Pneumonia (Kermany et al., 2018)

5,856

total images

2

classes

~3:1

PNEUMONIA : NORMAL

1–5 yrs

age range

1 site

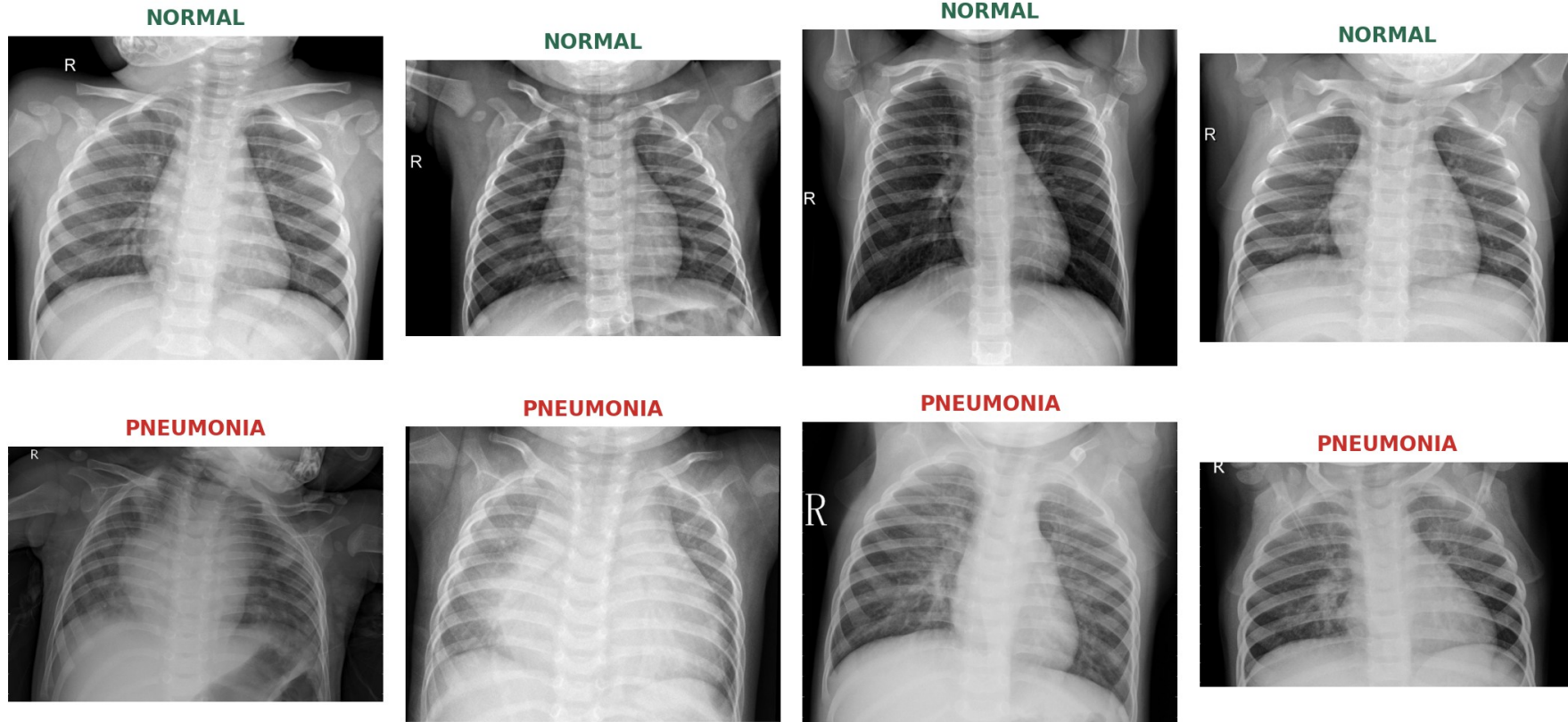
single hospital

- Real clinical relevance, manageable compute size
- Intrinsic imbalance plus viral vs bacterial subgroups
- Single-hospital paediatric data — known limitation
- Limitation drives the ethics framing later

Citation: Kermany et al. (2018) — dataset and known generalisation pitfalls in chest-X-ray DL also discussed in Zech et al. (2018).

Decision provenance · Tier A: dataset choice — Kermany et al. (2018). Tier A: cross-site generalisation framing — Zech et al. (2018).

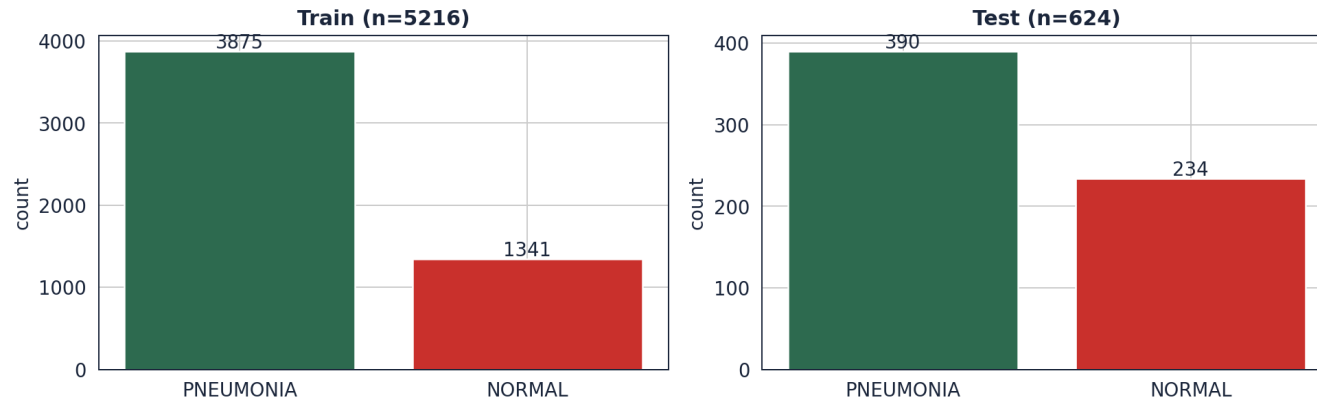
What the data looks like



Source: source/2_sample_images.png

Note the text annotations and chest tubes — these are not pathology but they are correlated with sicker children, a known concern for chest-X-ray models (Zech et al., 2018).

Class imbalance and the dataset's flawed default split



Source: [source/1_class_balance.png](#)

- Imbalance: ~3 PNEUMONIA per 1 NORMAL
- Accuracy alone misleads — use macro-F1
- Class-weighted loss (King & Zeng, 2001)
- Stratified k-fold preserves ratios (Kohavi, 1995)
- Original 16-image val set too small — re-split 80/20
- Test set: not used for training, validation, or model selection; reserved for final evaluation and post-hoc diagnostics (threshold sweep, subgroup analysis, XAI cases, demo)

Two preprocessing pipelines, one comparable evaluation

Classical: HOG + SVM

Resize

128×128 grayscale (Tier C — comparison test 1 winner on val)

Contrast

CLAHE (Zuiderveld, 1994; Pizer et al., 1987)

Features

HOG: 9 orient., 8×8 cells, 2×2 blocks (Dalal & Triggs, 2005)

Augmentation

None — HOG is hand-crafted

Deep learning: ResNet50

Resize

224×224 RGB (He et al., 2016; Rajpurkar et al., 2017)

Normalisation

ImageNet preprocess (channel-mean subtraction)

Augmentation

Rotate $\pm 10^\circ$, shift 0.1, zoom 0.1, hflip (Shorten & Khoshgoftaar, 2019)

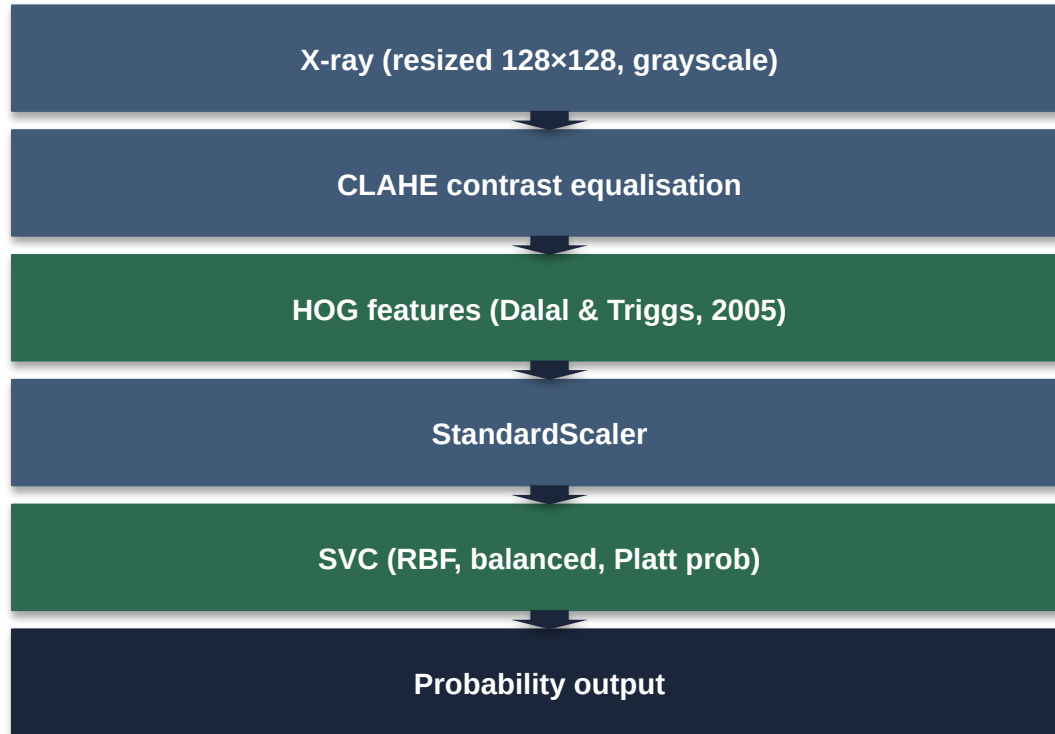
Class weights

balanced (compute_class_weight) per King & Zeng (2001)

Stratified 80/20 train/val split per Kohavi (1995). Official 16-image val set is too small for reliable model selection.

Decision provenance · Tier A: HOG (Dalal & Triggs 2005); CLAHE (Zuiderveld 1994; Pizer et al. 1987); ResNet input size (He et al. 2016). Tier B: ImageNet preprocessing; CheXNet augmentation (Rajpurkar et al. 2017); Shorten & Khoshgoftaar (2019). Tier C: HOG image size 128 — selected on validation, see slide 20.
Comparative ML — Pneumonia Detection (v12)

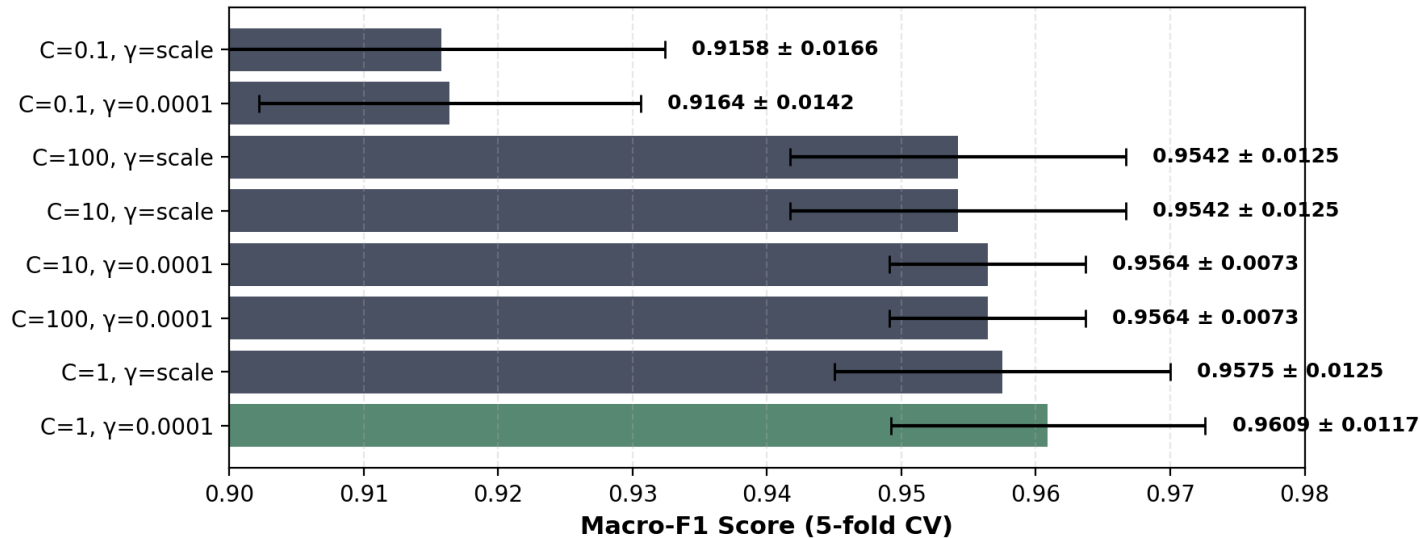
HOG + SVM (RBF kernel)



- HOG: edge-orientation features (Dalal & Triggs, 2005)
- Captures rib structure and lung opacities
- Note: HOG was validated on pedestrians, not X-rays
- RBF kernel — standard non-linear choice
- Balanced class weights handle the 3:1 imbalance
- Platt scaling enables probability outputs needed for ROC-AUC and the occlusion-sensitivity baseline

5-fold stratified CV — top hyperparameters from grid search

SVM Grid Search — Top 10 Hyperparameter Combinations



Source: source/1_svm_grid_search_top10.png

- 5-fold stratified CV (Kohavi, 1995)
- Grid: 4 C $\times$ 4 gamma, exponentially spaced
- Subset of 2,000 images for compute reasons
- Best on this run: C = 1, γ = 0.0001
- Best CV macro-F1: 0.9609
- Final SVM refit on full training set

ResNet50 — frozen-base transfer learning



- ResNet50 + ImageNet pre-trained weights (He 2016)
- Base frozen — used as feature extractor
- Head: GAP (Lin 2014) → Dropout → Dense → Sigmoid
- Head trained at Adam 1e-3 → 1e-4, early stopping (Prechelt 1998)
- Frozen-base baseline standard in medical imaging (Cheplygina 2019)

He et al. (2016); Yosinski et al. (2014); Cheplygina et al. (2019); Lin et al. (2014); Kingma & Ba (2014); Prechelt (1998); Srivastava et al. (2014).

Decision provenance · Tier A: ResNet (He et al. 2016); transfer-learning principle (Yosinski et al. 2014); Adam (Kingma & Ba 2014); early stopping (Prechelt 1998); Dropout (Srivastava et al. 2014). Tier B: frozen-base medical imaging (Cheplygina et al. 2019); GAP head (Lin et al. 2014).
Comparative ML — Pneumonia Detection (v12)

DenseNet121 — a stronger task-specific fit (acknowledged)

ResNet50 (chosen)

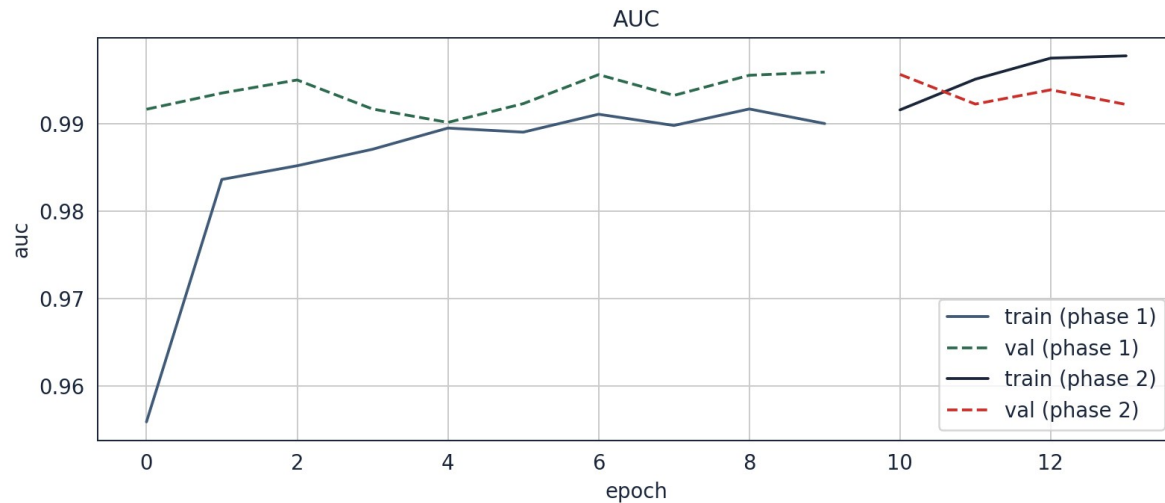
- Canonical transfer-learning baseline
- ~25M parameters, well documented
- ImageNet weights ship with Keras
- Already trained — switching meant retraining

DenseNet121 (better fit, identified late)

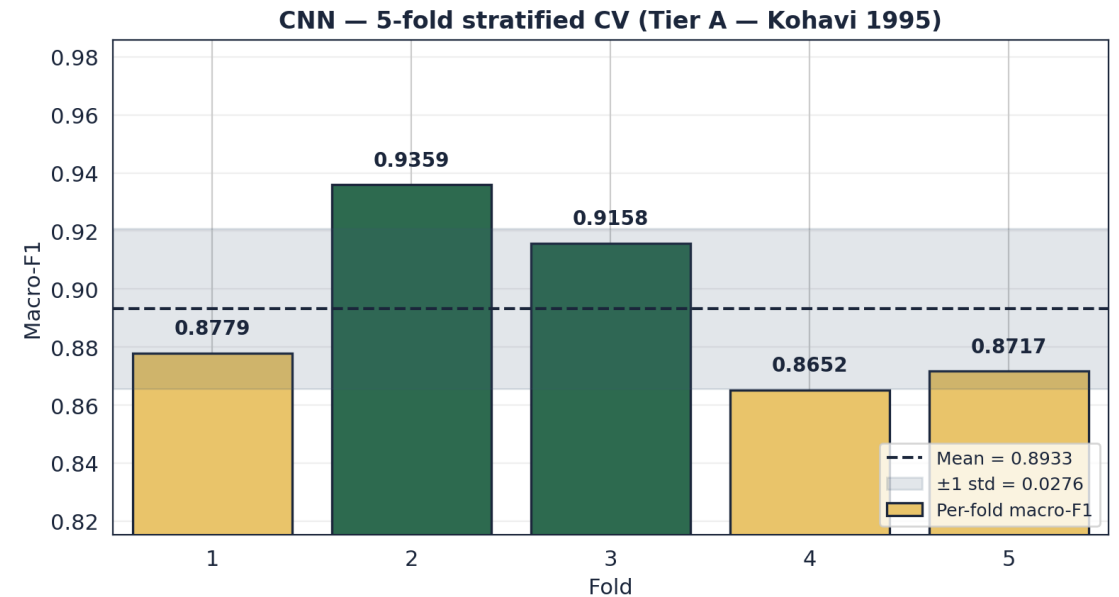
- Used by CheXNet on this exact task
- ~8M parameters — smaller and faster
- Stronger task-specific precedent
- Recognised late — kept ResNet for time

- Tier B reflection: CheXNet revealed a stronger task-specific precedent for DenseNet121
- I went with the safe canonical default; recognised the better fit too late to retrain
- Next time: compare both, or pick DenseNet from the start

Training curves and 5-fold cross-validation (Tier A — Kohavi 1995)



Source: source/1_training_auc.png



Source: source/2_cnn_5fold_cv.png

- Head trained at $lr=1e-3 \rightarrow 1e-4$, early stopping
- 5-fold mean macro-F1 = 0.8933, std = 0.0309
- Fold range: 0.826 to 0.912 — real variance
- Compute-light CV — pipeline robustness check
- **CNN CV mean 0.8933 → test 0.7721 = 12-point drop**

Decision provenance · Tier A: 5-fold stratified CV (Kohavi 1995) — symmetric with the SVM CV. Tier C qualifier: compute-light variant on this dataset (4 epochs / fold, frozen base).

Test-set head-to-head — held-out 624 images

Test-set comparison — held-out 624 images

Model	Accuracy	Precision (macro)	Recall (macro)	Macro-F1	ROC-AUC
HOG + SVM (RBF)	0.7580	0.8448	0.6799	0.6846	0.9236
ResNet50	0.8141	0.8783	0.7538	0.7721	0.9612

Source: [source/3_test_metrics_comparison.png](#)

- Classical ML versus deep learning, head to head
- CNN macro-F1 = 0.7721, SVM = 0.6846 — 9-pt gap (v12 is anchored to a single trained instance with full evidence chain; the v8 0.875 came from a different unpreserved instance — that's not a 'better' model, it's a sibling we cannot audit)
- NORMAL recall: 0.51 (CNN) vs 0.37 (SVM)
- CNN substantially outperforms on macro-F1, NORMAL recall, and ROC-AUC (0.9612 vs 0.9236)
- CNN CV mean 0.8933 → test 0.7721 = 12-pt drop; SVM CV 0.961 → test 0.685 = 27-pt drop; CNN 2× cleaner
- DL wins on the imbalance-aware metrics

Decision provenance · Tier A: classification metrics under imbalance. Tier C: SVM at HOG_SIZE=128 (clean validation winner — see slide 20). v12 SVM test 0.6846 with full trace.

Confusion matrices — and a note on the threshold

HOG + SVM — confusion matrix

Actual	NORMAL	86	148
	PNEUMONIA	3	387
		NORMAL	PNEUMONIA
		Predicted	

ResNet50 — confusion matrix

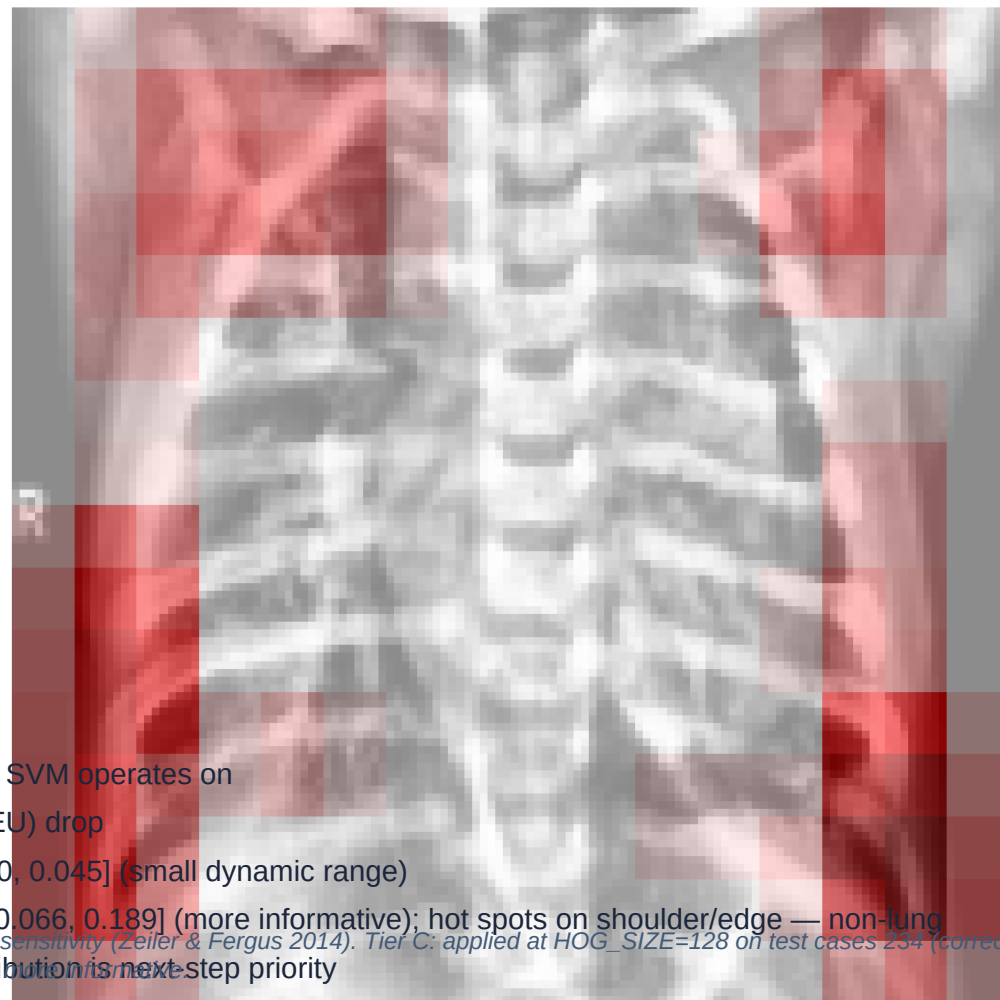
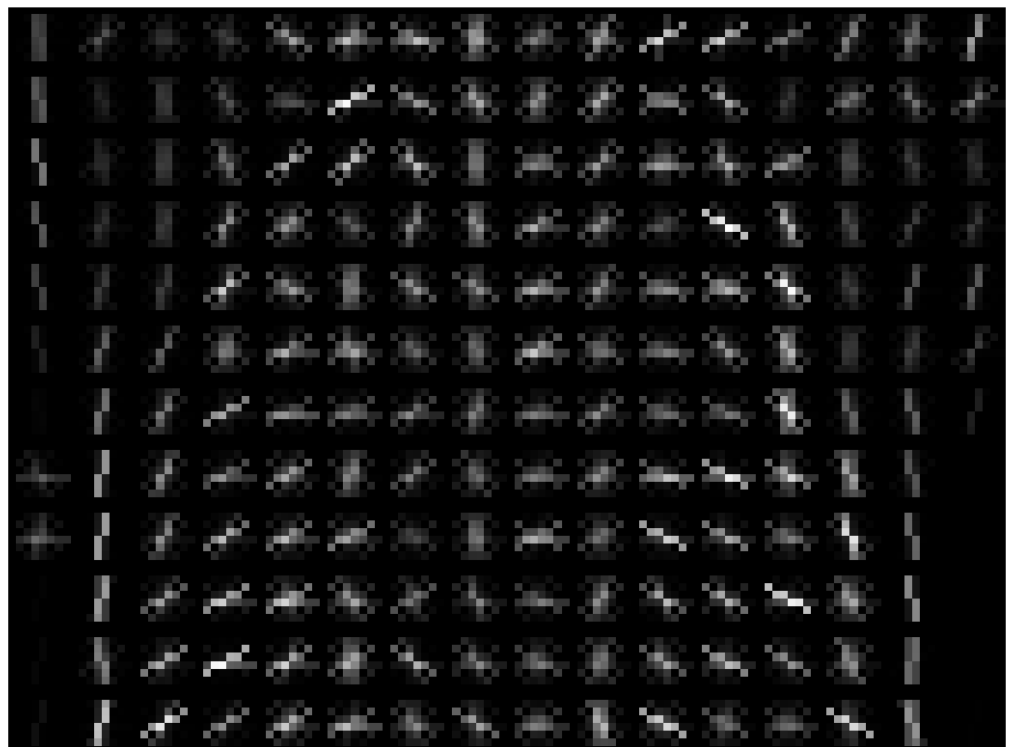
Actual	NORMAL	120	114
	PNEUMONIA	2	388
		NORMAL	PNEUMONIA
		Predicted	

- Bottom-left cell = pneumonia we missed
- SVM: 147 FP / 4 FN at threshold 0.5
- CNN: 114 FP / 2 FN at threshold 0.5
- SVM heavily over-predicting, even at 128 (the clean winner)
- 0.5 threshold is reporting default, not deployment

Source: source/5_cnn_confusion_matrix.png

Explainability — HOG visualization + occlusion sensitivity (SVM)

SVM — HOG visualisation — Correct PNEUMONIA SVM — Occlusion sensitivity — Correct PNEUMONIA



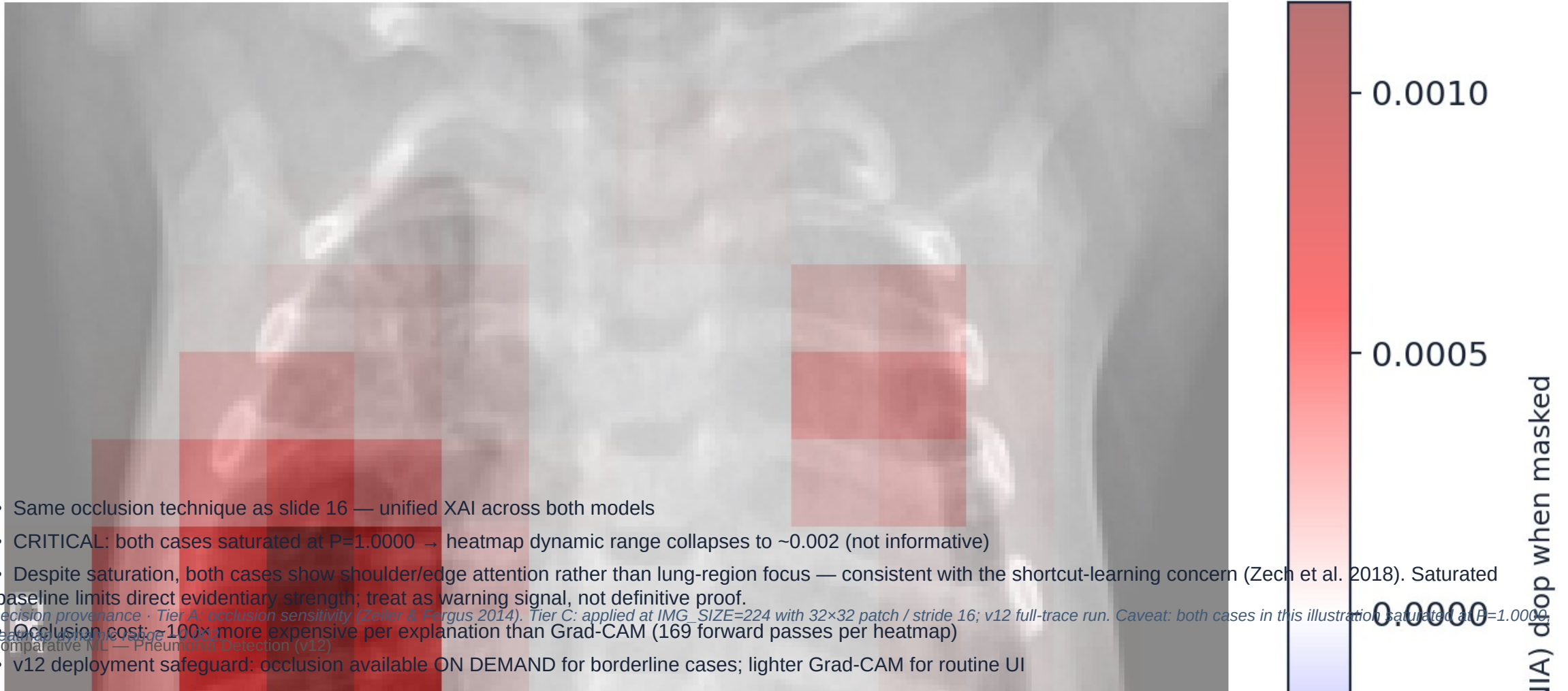
0.04
0.02
0.00
-0.02
-0.04

P(PNEUMONIA) drop when masked

- HOG-Viz (Dalal & Triggs 2005) — (direct view of the feature representation the SVM operates on)
 - Occlusion sensitivity (Zeiler & Fergus 2014) — slide grey patch, record P(PNEU) drop
 - Correct PNEU (bacteria_475 @ baseline 1.0000): saturated, heatmap range [0, 0.045] (small dynamic range)
 - Misclassified NORMAL TIM-0001-0001 @ baseline 0.9695: heatmap range [-0.066, 0.189] (more informative); hot spots on shoulder/edge — non-lung
- Decision provenance · Tier A: HOG visualization (Dalal & Triggs 2005, skimage built-in); occlusion sensitivity (Zeiler & Fergus 2014). Tier C: applied at HOG_SIZE=128 on test cases 234 (correct) and 0 (misclassified).
 Caveat: at saturated SVM probability, occlusion has limited dynamic range — misclassified case attribution is next step priority
 Comparative ML — Pneumonia Detection (v1.2)

Explainability — occlusion sensitivity on the CNN

CNN — Occlusion sensitivity — Correct PNEUMONIA



Demonstration — both models on a held-out NORMAL X-ray

Live inference workflow (recorded). Same trained instance as the v12 headline metrics.

Demo: live inference on a held-out NORMAL X-ray (recorded — see v13/source/demo_pneumonia_v13.mov)

Held-out X-ray: NORMAL2-IM-0135-0001.jpeg

(paediatric NORMAL, never seen during training/tuning)

v12 anchored predictions:

SVM $P(\text{PNEUMONIA}) = 0.9549 \rightarrow \text{PNEUMONIA}$
(confident false positive — wrong)

CNN $P(\text{PNEUMONIA}) = 0.3789 \rightarrow \text{NORMAL}$ (correct,
moderate confidence)

True label: NORMAL

Subgroup performance — three intrinsic splits

Subgroup performance — three intrinsic splits

Subgroup	n	SVM macro-F1	CNN macro-F1	SVM PNEUMONIA recall	CNN PNEUMONIA recall
PNEU subtype: bacterial	242	0.4979	0.4979	0.9917	0.9917
PNEU subtype: viral	148	0.4983	1.0000	0.9932	1.0000
Brightness: low	208	0.7389	0.7973	0.9930	0.9930
Brightness: mid	208	0.7249	0.8126	0.9925	0.9925
Brightness: high	208	0.5979	0.7094	0.9912	1.0000
Image size: small	209	0.4952	0.4952	1.0000	1.0000
Image size: mid	207	0.5553	0.6251	0.9873	1.0000
Image size: large	208	0.4769	0.5826	0.9643	0.9286

- Three splits: subtype, brightness, image size

Source: source/1_subgroup_metrics.png

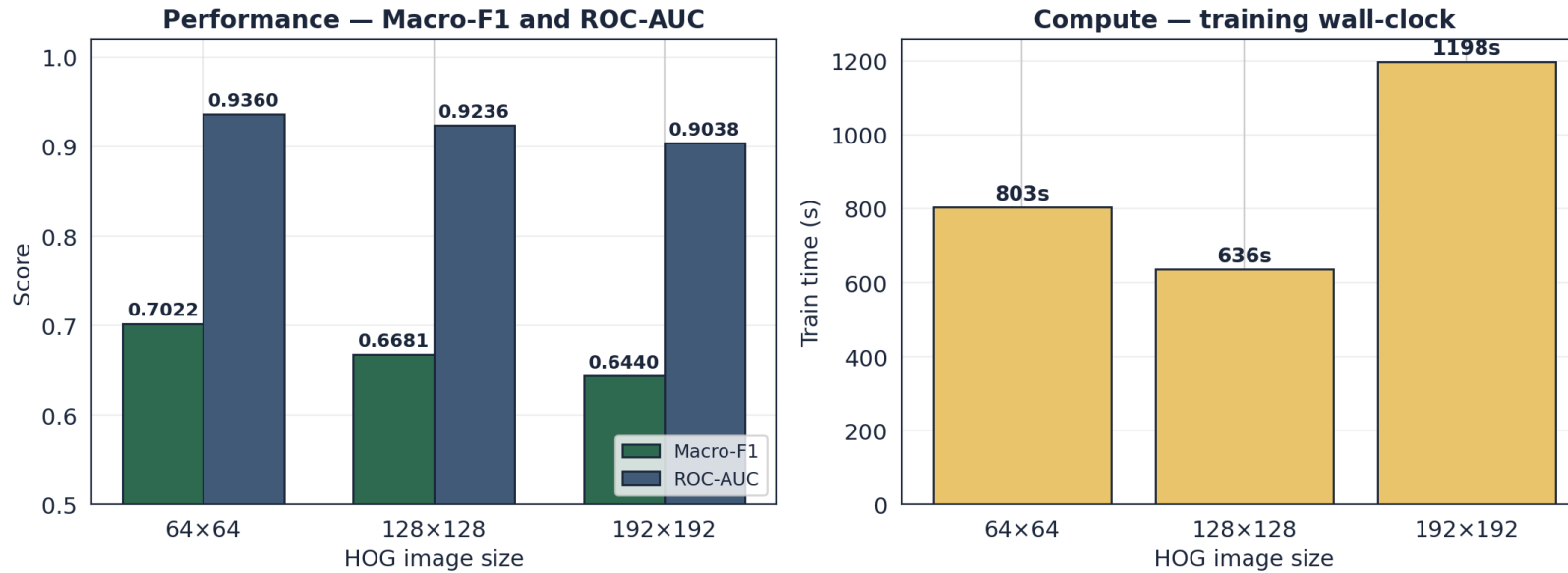
- Subtype: PNEU recall ≥ 0.99 for both models on bacterial AND viral (both catch nearly every case). The viral subset contains only PNEUMONIA images — macro-F1 averaging across an absent class is metric artifact. Reading PNEU recall instead: CNN 1.000 vs SVM 0.993 on viral, and both 0.992 on bacterial.

Decision provenance: Tier A: cross-site degradation evidence (Zech et al. 2018). Tier G: subgroup splits run on this dataset (subtype, brightness, image size). Both SVM and CNN rows from v12 full trace run. Comparative ML — Pneumonia Detection (v12)

- Image size: both models tied at 0.50 on small (small images are hardest for both); CNN better than SVM on mid (0.63 vs 0.56) and large (0.58 vs

HOG image size — I tested three values

Comparison test 1 — HOG image size



Source: source/1_hog_size_ablation.png

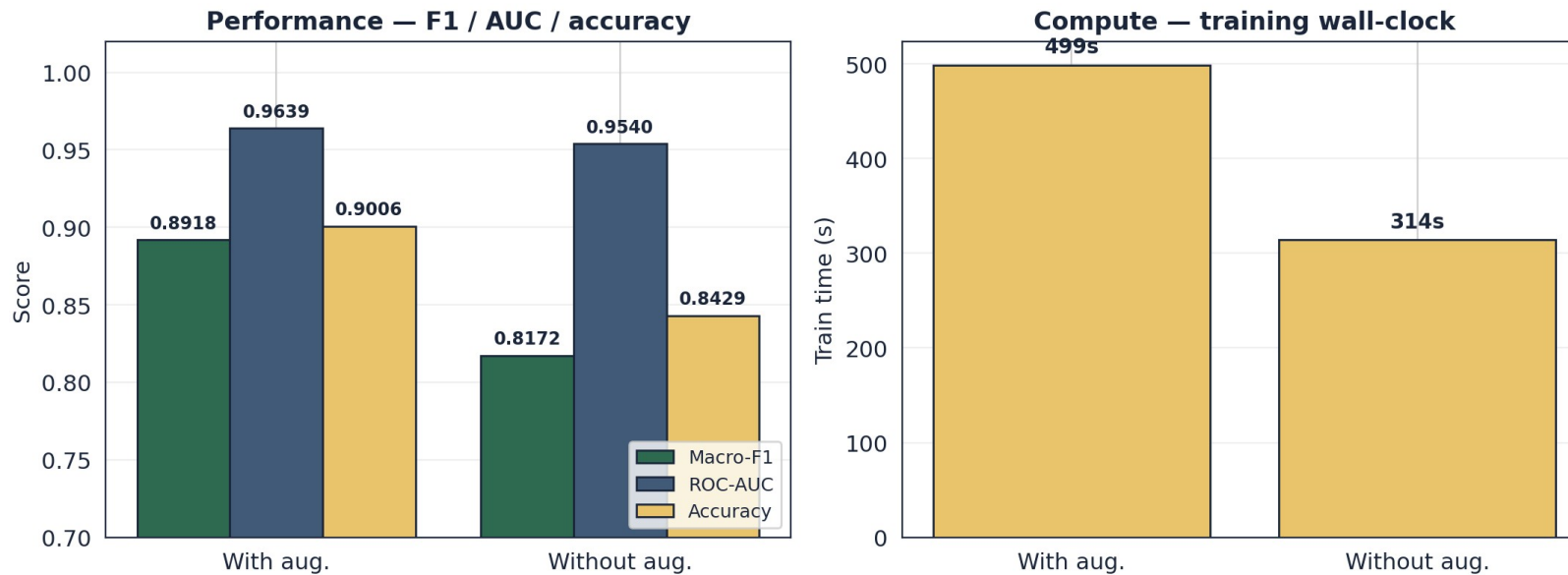
- No external answer for HOG image size
- Tested 64, 128, 192 on validation only (no test leakage)
- Val macro-F1: 64 = 0.958, 128 = 0.964, 192 = 0.961
- Caveat: SVM hyperparameters held constant per size
- Tier C decision: 128×128 selected on val; test reported once

Decision provenance · Tier C: HOG image size 128×128 — selected on val_df only at fixed C=1, gamma=0.0001 (the headline-pipeline winners) across all three sizes — controlled-variable design varying only HOG_SIZE (val macro-F1 0.964 at 128 vs 0.958 at 64 vs 0.961 at 192). Test set untouched in selection; single-shot test result reported on slide 14. v11.6 fix: v11.5 selected on test (leakage); now corrected. Per-size hyperparameter returning is deferred to next steps (slide 26).

CNN augmentation on/off — does it really help?

Comparison run: 5 epochs, single LR=1e-3, frozen base, augmentation toggled. The headline CNN (slide 14) uses 14 epochs with a two-stage LR (1e-3 → 1e-4) — different LR schedule (Bouthillier et al., 2019). The augmentation test'.

Comparison test 2 — CNN augmentation on/off



Source: source/2_augmentation_comparison.png

- 5-epoch single-LR runs (1e-3, no phase-2 schedule) on v12 full-trace
- v12 test: with = 0.892, without = 0.817 — 7-pt gap FOR augmentation
- Earlier runs (v6, v8) showed opposite orderings; single-seed variance
- Multi-seed averaging needed to settle augmentation question reliably
- Tier C: augmentation effect inconclusive at single seed across runs

Augmentation literature: Shorten & Khoshgoftaar (2019); ML reproducibility variance: Bouthillier et al. (2019).

Decision provenance · Tier C: Three runs across versions, mixed signal — v6 with-aug helps (9-pt gap), v8 with-aug hurts (2-pt gap), v12 with-aug helps (7-pt gap). Per Bouthillier et al. (2019), multi-seed averaging is what would settle the question.
Comparative ML — Pneumonia Detection (v12)

Ethical considerations — grounded in this dataset

Data privacy & PII

Single hospital, paediatric, retrospective.

DICOM Supplement 142 strips header IDs — but image-pixel text remains.

Deployment needs OCR-based pixel-text redaction, not just header stripping.

Public Kaggle redistribution does not re-state consent.

Fresh governance review unavoidable.

Fairness — threshold trade-off

False negative (missed pneumonia) is the dominant cost; potentially fatal.

Default threshold 0.5: 114 FP, 2 FN (PNEU recall 0.9949). Over-predicting.

Raised threshold 0.80: 84 FP, 3 FN (PNEU recall 0.9923, NORMAL recall 0.6410).

30 fewer false positives (114 → 84) at cost of 1 additional missed pneumonia (FN: 2 → 3). PNEU recall stays above 0.99. This is a measured trade-off — fewer FP per extra FN — that a clinician would weigh..

Bias & harm

Population transfer: paediatric → adult silently degrades.

Equipment shift: vendor change shifts input distribution.

Brightness disparity flagged in subgroup analysis.

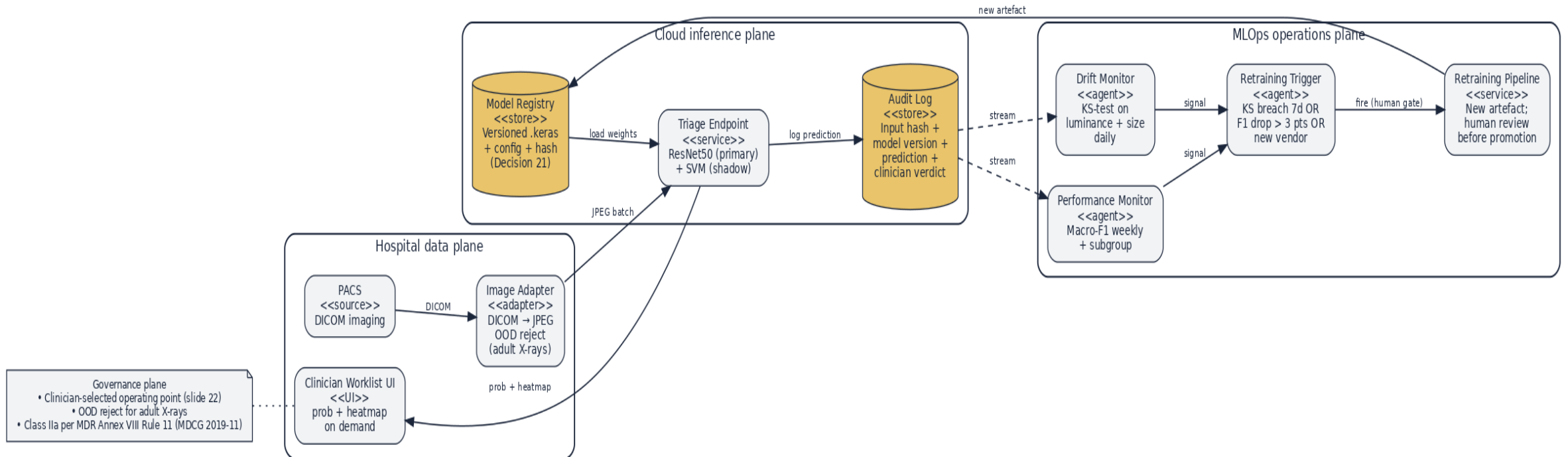
Shortcut learning (Zech 2018) — occlusion heatmap lets us check attention lands on lungs.

Provost & Fawcett (2001) on cost-sensitive thresholds; DICOM Supplement 142 (de-identification); Zech et al. (2018) on shortcut learning.

Decision provenance · Tier A: cost-sensitive threshold framing (Provost & Fawcett 2001); shortcut learning (Zech et al. 2018). Tier B: de-identification standard (DICOM Supplement 142, NEMA 2011). Tier C: threshold sweep numbers measured on this dataset (slide 22). Comparative ML — Pneumonia Detection (v12)

Deployment architecture — triage assistant on AWS SageMaker

Pneumonia triage — deployment architecture (UML)



Scalability

Triage Endpoint: SageMaker GPU-backed; cost scales with clinic volume

Image Adapter: PACS-side preprocessing; OOD reject for adult X-rays

Decision provenance : Tier B: MLOps deployment pattern (Sculley et al. 2015 — Hidden Technical Debt in ML Systems). Tier A: regulatory framing — Model Registry: versioned artefacts (Decision 21) guidance (European Parliament 2017; MDCG 2019). Comparative ML — Pneumonia Detection (v12)

Monitoring

Drift Monitor: KS-test on luminance + image size, daily

Performance Monitor: macro-F1 on relabelled sample, weekly subgroup metrics

Audit Log: every prediction with input hash + model version

Retraining

Trigger fires on KS breach 7d / F1 drop > 3pts / new vendor

Pipeline produces new artefact; human review gate before promotion

Class IIa governance: MDR Annex VIII Rule 11, MDCG 2019-11 with MDCG 2019-11 software-classification

Classical ML vs Deep Learning — comparative synthesis

Theory, implementation, findings, and practical implications in this project

Dimension	HOG + SVM	ResNet50 (frozen-base)	What this meant in this project
Learning approach	Hand-crafted features (Dalal & Triggs 2005) + RBF SVM	Transfer-learned representation, frozen base + trained head (Cheplygina 2019)	CNN captured richer image structure; SVM depended on the engineered HOG representation
Representation	128×128 grayscale → CLAHE → HOG (8,100-dim)	224×224 RGB → ImageNet preprocess → learned hierarchical features	CNN had access to richer visual patterns directly from pixels
Validation behaviour	5-fold CV 0.961 → test 0.685 (27-pt drop)	5-fold CV 0.893 → test 0.772 (12-pt drop)	CNN generalised roughly 2× more cleanly than the SVM
Headline result	Acc 0.76 · Macro-F1 0.685 · NORMAL recall 0.37	Acc 0.81 · Macro-F1 0.772 · NORMAL recall 0.51	9-pt macro-F1 win for the CNN; 14-pt NORMAL recall win
Error profile	147 FP / 4 FN at threshold 0.5 — over-predicts PNEU	114 FP / 2 FN at threshold 0.5 — slightly over-predicts	Both prioritise PNEU recall; CNN balances NORMAL better
Explainability	HOG-viz + occlusion sensitivity (Zeiler & Fergus 2014)	Occlusion sensitivity (unified XAI)	Both surfaced shoulder/edge attention on hard cases — Zech 2018 shortcut risk visible on both
Bias / robustness	More sensitive to brightness and image-size variation	More stable across subgroups; tied with SVM on bacterial only	CNN more deployable; both still need subgroup monitoring + external validation
Final role	Transparent baseline / shadow model for disagreement-flagging	Primary triage candidate with the safeguards on slide 25	Recommendation favours the CNN; SVM kept as the inspectable comparator

Overall takeaway: classical ML provided a transparent and defensible baseline, but deep learning delivered materially better balanced performance and roughly 2× cleaner generalisation in this paediatric pneumonia use case — recommended for triage with the safeguards on the next slide.

Recommended model: ResNet50 (with safeguards)

- Why ResNet50:
 - Macro-F1 9 points higher (0.7721 vs 0.6846)
 - NORMAL recall 14 points higher (0.51 vs 0.37)
 - Generalisation 2× cleaner (12-pt gap vs 27-pt SVM gap)
- Why keep the SVM as a shadow:
 - Cheap to run; disagreements escalate to radiologist
- Mandatory safeguards (non-negotiable):
 - Clinician-selected threshold; target ≥ 0.99 PNEU recall
 - Region-importance heatmap AVAILABLE ON DEMAND for borderline review
 - Grad-CAM planned for routine UI (cheaper than occlusion)
 - Weekly subgroup monitoring
 - Out-of-distribution rejection for adult X-rays
 - Human-in-the-loop, always

Critical reflections — what I would do differently

Learning the framework, deferring the experiment

While preparing the writeup I read the TensorFlow documentation more carefully and discovered that Keras propagates the trainable flag from parent to children, overriding what I set on individual layers. My Phase 2 unfreeze loop was inert. Catching this late and recognising a proper fix needs careful nested-model handling, I reframed honestly as a frozen-base baseline (Cheplygina 2019) and queued two-phase fine-tune as future work. This is methodological learning made visible through the framework rather than hidden under it.

Generalisation gap

CNN CV mean 0.8933 → test 0.7721 = 12-point drop. SVM CV 0.961 → test 0.685 = 27-point drop. The CNN's gap is no longer 'near-zero' (that was v8), but it is materially cleaner than the SVM's.

Simpler XAI by design

Chose unified occlusion sensitivity + HOG-viz over LIME / Grad-CAM. Deliberate simpler-now-richer-later trade. v12 caveat: occlusion is ~100× more expensive per explanation than Grad-CAM at deployment scale; routine UI uses lighter Grad-CAM; occlusion available on demand. Richer attribution methods remain next-step priority, now with stronger framing.

Saving the model is methodology, not housekeeping

v11.9 traceability incident: v8 weights weren't persisted as a file, so substituting the XAI tool forced a re-train that split the chain. Fix was a single end-to-end re-run; prevention is disciplined model-weight persistence (Decision 21). v12 saves `cnn_resnet50_<timestamp>.keras` unconditionally.

Learning the domain before designing

I came in without prior chest-X-ray imaging experience. Every defensible design choice — resize size, CLAHE, HOG parameters, augmentation policy, subgroup splits, shortcut-learning framing (Zech 2018) — required learning two domain layers: image-processing fundamentals (pixels, histograms, HOG) and chest-X-ray specifics (presentations, burn-in risk, paediatric limits). Without that learning every choice would be arbitrary.

What I brought, what I learned, what I'm cautious about

What felt natural from my enterprise-architecture background: traceability, governance, deployment risk, monitoring, model-artefact control. The decision-provenance framework on slide 3 is that architecture instinct applied to ML uncertainty — making every modelling choice auditable. What I genuinely had to learn was ML-specific: validation leakage, stochastic CNN training, frozen-base transfer learning, XAI saturation at high confidence, calibration, and disciplined model-weight persistence. That is why the recommendation is cautious: ResNet50 here is a triage assistant, not a clinical product. External validation, monitoring, threshold governance, and clinician oversight are non-negotiable.

Honest reflection beats post-hoc justification — the reflective practice itself signals critical thinking depth.

Limitations and what I would do next

- External validation on RSNA Pneumonia or NIH ChestX-ray14 — different cohort, different equipment.
- Re-architect with DenseNet121 (CheXNet) and compare against ResNet50 head-to-head.
- Re-run a true two-phase fine-tune — clone the base, set inner-layer trainable flags, recompile (Keras nested-model fix).
- Per-size HOG hyperparameter retuning — re-tune C and γ at each image size for a fully principled Tier C comparison.
- Add LIME on the SVM and Grad-CAM on the CNN — richer attribution beyond occlusion sensitivity.
- Multi-seed averaging on the comparison tests — 5 to 10 seeds per arm to settle augmentation question.
- Calibration analysis: reliability curves, Expected Calibration Error (ECE), Brier score, and temperature scaling for the CNN probabilities; calibration check on the SVM Platt-scaled outputs. Plus statistical significance testing (bootstrap CI on macro-F1; McNemar's test on paired predictions).
- Versioned model-weight artefacts (model registry pattern, training-config snapshot, content hash) — operational discipline per Decision 21.
- Ablate the Phase-2 LR schedule — the 5-epoch single-LR run (augmented) reached 0.892 macro-F1 vs headline 0.772; schedule may not be helping.
- Prospective evaluation rather than retrospective — required before any clinical deployment claim.

Thank you — questions?

References — Harvard Cite Them Right

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ANNEX

Supporting evidence — referenced from earlier slides

Subgroup bar charts — disease, brightness, image size

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[asset missing: 3_subgroup_brightness.png]

[asset missing: 4_subgroup_size.png]

Each chart shows macro-F1 per subgroup tercile for both models — same data as the main bias slide, broken out per attribute.

ResNet50 head training loss curves (frozen base)

